IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM Granite and IBM Bob”

Project Tile – [Agentic AI Health Symptom Checker]

Domain of Project – [HealthCare]

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Problem Statement -

Problem Statement : Agentic AI Health Symptom Checker

The Challenge
People often struggle to understand their symptoms and determine when medical attention is needed. Unreliable online health information can also lead to misinformation and unnecessary anxiety.

The Objective
Build an Agentic AI Health Symptom Checker that provides personalized and reliable health guidance. The solution should:

Symptom Analysis – Analyze user symptoms and identify possible health conditions.

Medical RAG Retrieval – Retrieve verified medical information from trusted sources.

Health Risk Assessment – Assess urgency and identify potential warning signs.

Preventive Health Guidance – Provide self-care advice and when to consult a doctor.

Multi-Agent System –
Symptom Agent (analyzes symptoms)
Medical Knowledge Agent (retrieves medical data)
Health Advisory Agent (provides health guidance)
Safety Agent (checks risks and red flags)

Multi-Language Support – Support symptom input and guidance in multiple languages.

Health Dashboard – Display symptom history, risk levels, and recommendations.

Proposed Solution -

Proposed Solution: AI-Powered Nutrition Agent
The proposed solution is an Agentic AI Health Symptom Checker built using IBM Cloud, watsonx.ai, and IBM Granite 4 to provide personalized, evidence-based, and safe health guidance.

The system analyzes user-provided symptoms along with age, duration, severity, existing conditions, medications, and allergies to provide relevant health insights. It retrieves verified medical information from trusted sources such as WHO, CDC, and government health guidelines to support its responses.

The solution consists of four key agents:

Symptom Analysis Agent – Understands and analyzes symptoms and patient context.
Medical Knowledge Agent – Retrieves relevant and verified medical information.
Health Advisory Agent – Provides preventive advice, home-care guidance, and recommended next steps.
Safety & Triage Agent – Detects red-flag symptoms and determines the appropriate urgency level.

The platform provides an interactive dashboard showing possible conditions, urgency level, home remedies, warning signs, and when to consult a doctor, along with multi-language support.

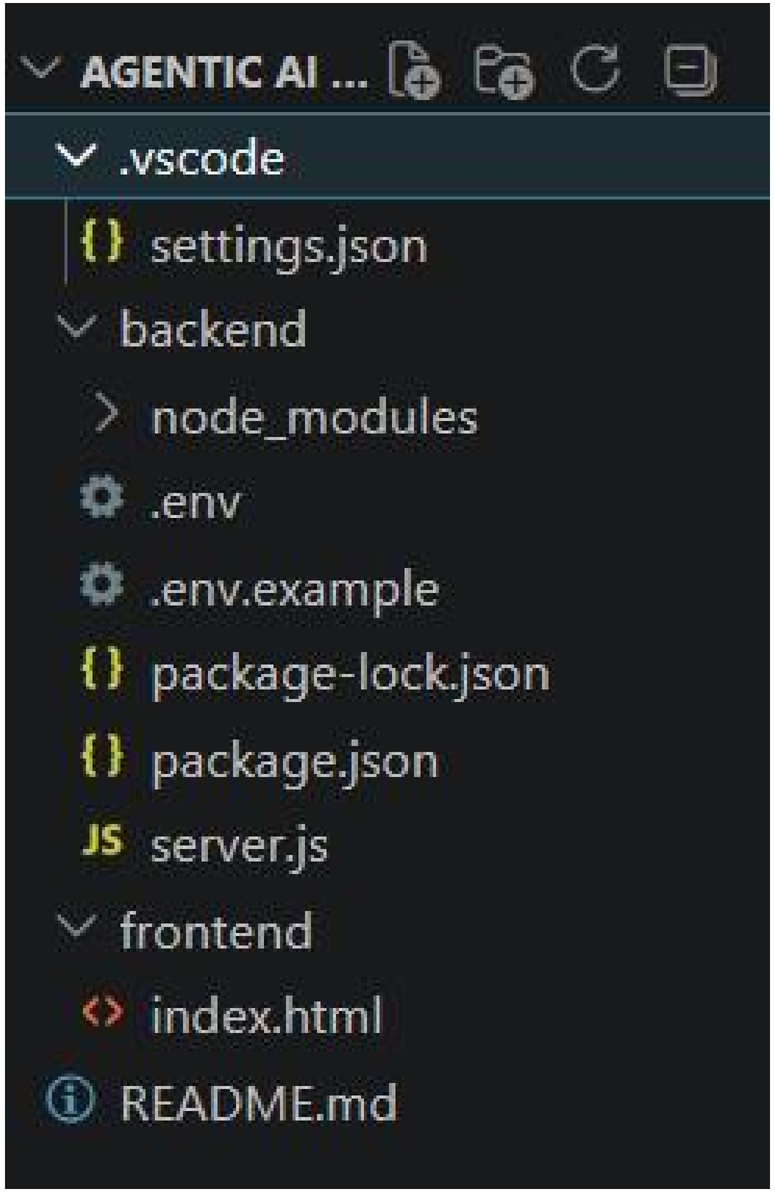
Overall, the solution provides a safe, personalized, and accessible AI health assistant powered by IBM Granite.

Technology Used

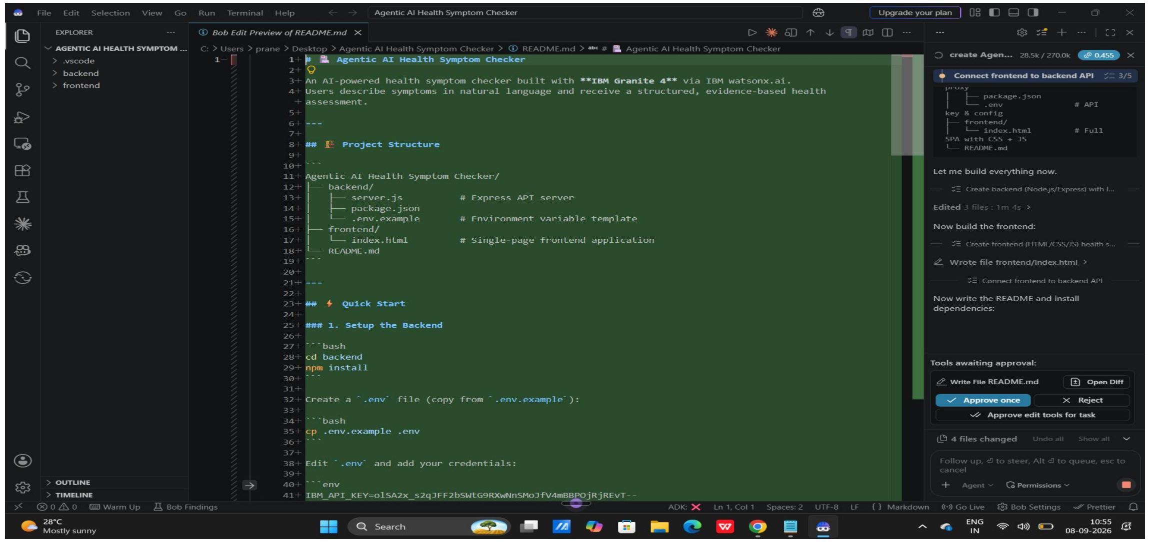
- IBM Bob Platform – For designing and orchestrating multi-agent workflows visually.
- IBM Granite Model – e.g., ibm-granite-3-2-8b – For natural language understanding, symptom analysis, reasoning, and personalized health guidance.
- IBM Cloud – Provides the infrastructure to build, deploy, and scale the Health Symptom Checker securely and efficiently.
- Medical Knowledge Retrieval – Fetches verified medical information from trusted sources such as WHO, CDC, and government health guidelines to support accurate responses.
- IBM watsonx.ai – For model deployment, AI development, monitoring, and governance.
- Vector Database (FAISS/Chroma) – Only if you are actually using RAG to store and retrieve medical documents efficiently.

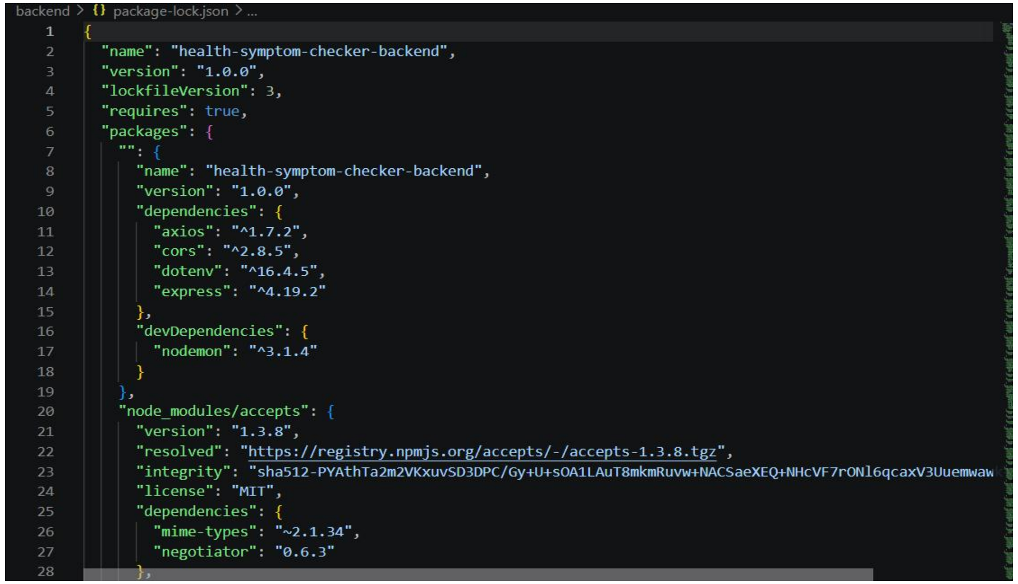
Required Files created using IBM bob -

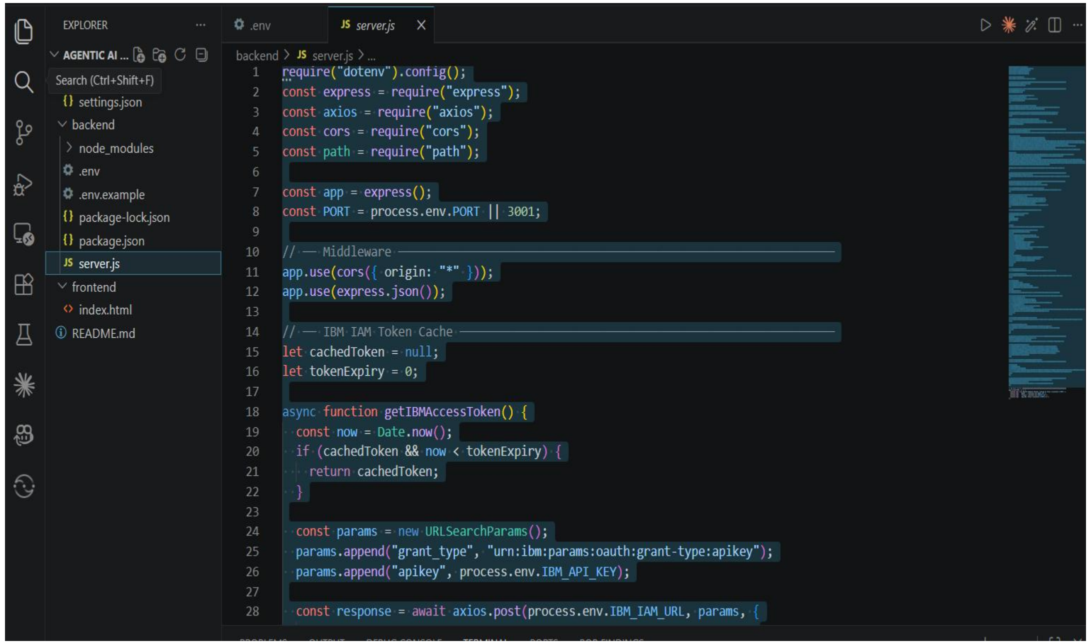
```
backend/
├── server.js
├── package.json
├── .env
└── .env.example

frontend/
├── index.html
├── style.css
└── script.js

├── README.md
└── .gitignore
```

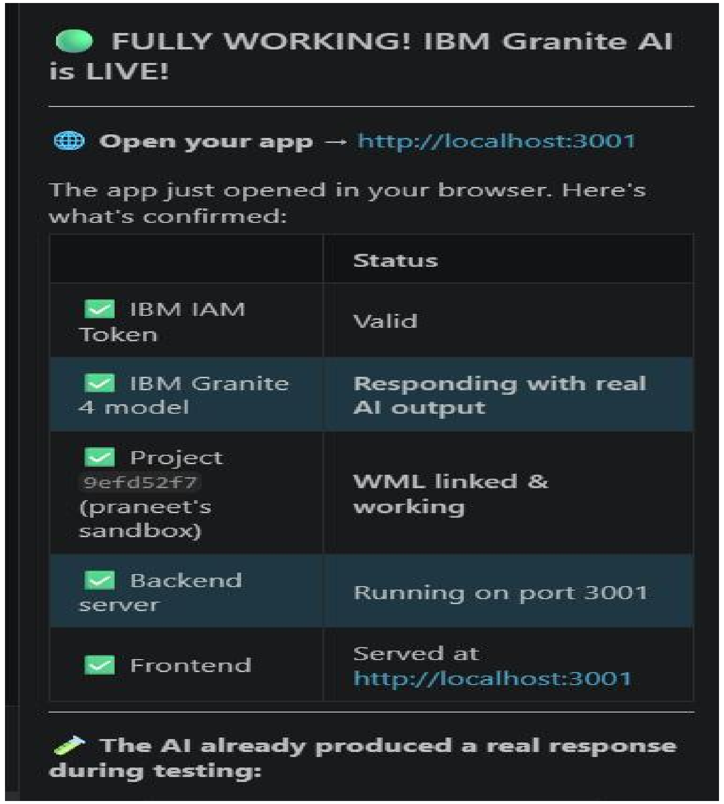


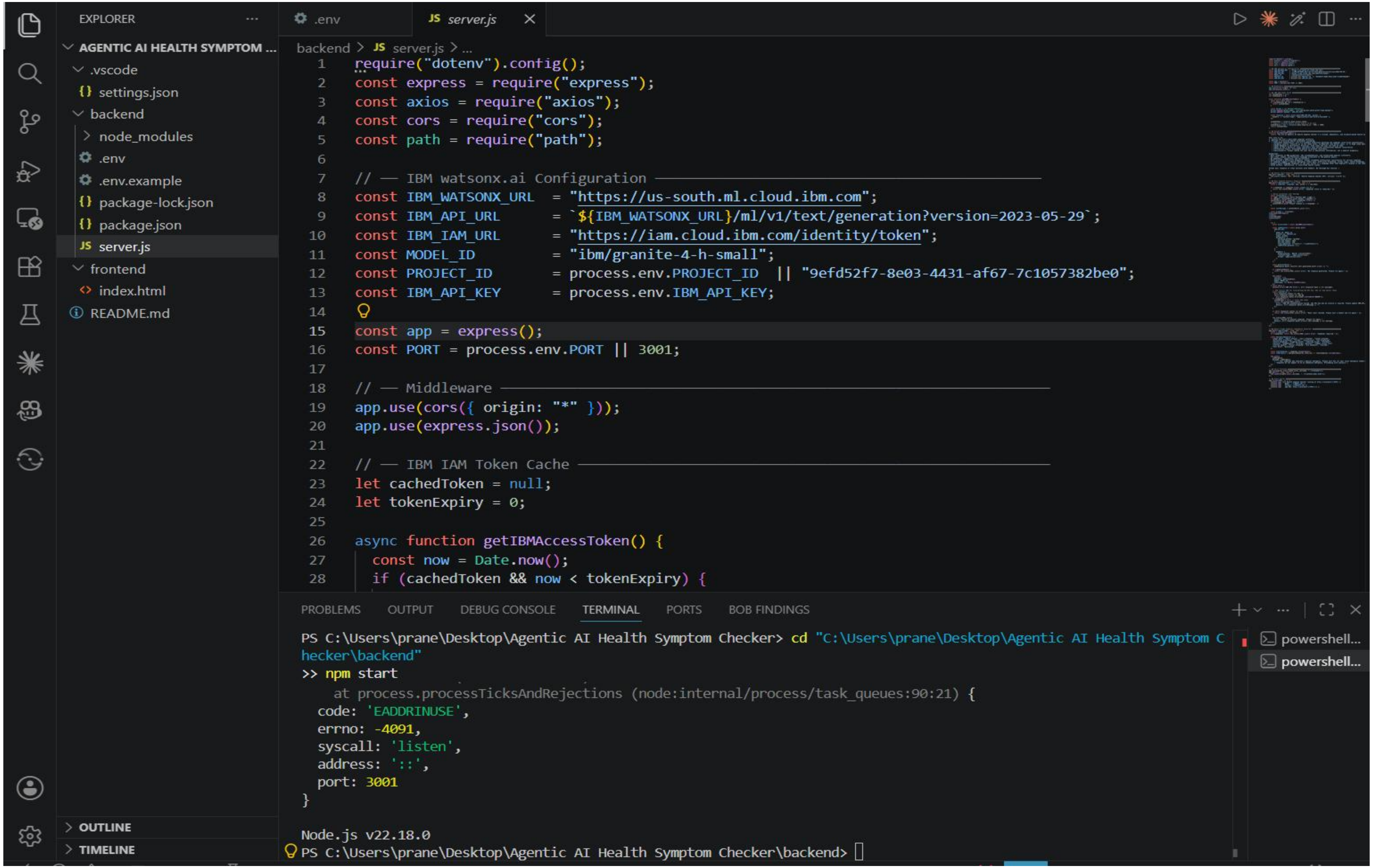
IBM Bob Screenshots:



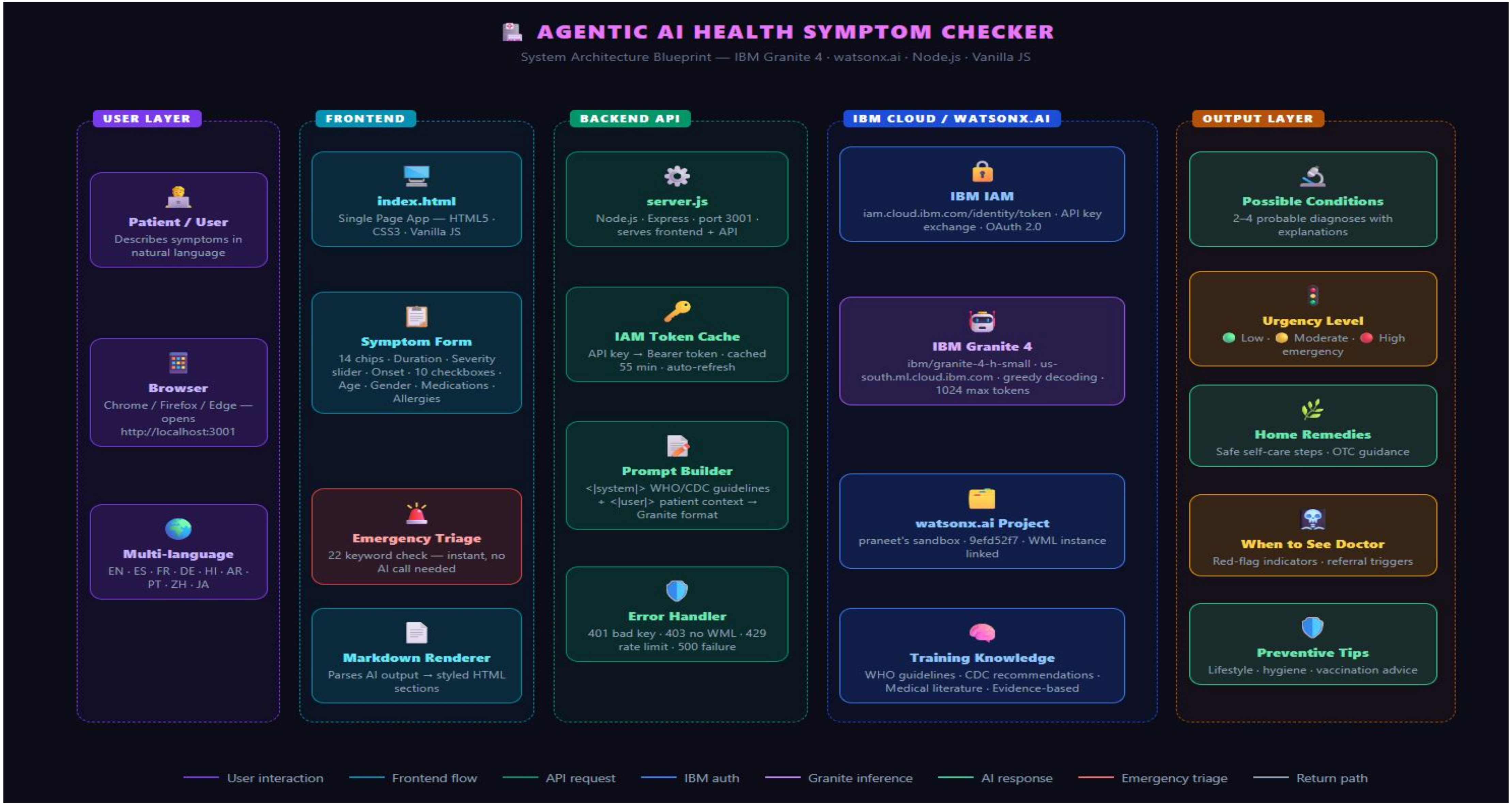







EXPLORER
.env
server.js
AGENTIC AI HEALTH SYMPTOM ...
.vscode
settings.json
backend
node_modules
.env
.env.example
package-lock.json
package.json
server.js
frontend
index.html
README.md
OUTLINE
TIMELINE
backend > server.js > ...
require("dotenv").config();
const express = require("express");
const axios = require("axios");
const cors = require("cors");
const path = require("path");
// ── IBM watsonx.ai Configuration ──
const IBM_WATSONX_URL = "https://us-south.ml.cloud.ibm.com";
const IBM_API_URL = `${IBM_WATSONX_URL}/ml/v1/text/generation?version=2023-05-29`;
const IBM_IAM_URL = "https://iam.cloud.ibm.com/identity/token";
const MODEL_ID = "ibm/granite-4-h-small";
const PROJECT_ID = process.env.PROJECT_ID || "9efd52f7-8e03-4431-af67-7c1057382be0";
const IBM_API_KEY = process.env.IBM_API_KEY;
const app = express();
const PORT = process.env.PORT || 3001;
// ── Middleware ──
app.use(cors({ origin: "*" }));
app.use(express.json());
// ── IBM IAM Token Cache ──
let cachedToken = null;
let tokenExpiry = 0;
async function getIBMAccessToken() {
const now = Date.now();
if (cachedToken && now < tokenExpiry) {
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
BOB FINDINGS
PS C:\Users\prane\Desktop\Agentic AI Health Symptom Checker> cd "C:\Users\prane\Desktop\Agentic AI Health Symptom Checker\backend"
>> npm start
at process.processTicksAndRejections (node:internal/process/task_queues:90:21) {
code: 'EADDRINUSE',
errno: -4091,
syscall: 'listen',
address: '::',
port: 3001
}
Node.js v22.18.0
PS C:\Users\prane\Desktop\Agentic AI Health Symptom Checker\backend>
powershell...
powershell...

Architecture blueprint for IBM bob:



Role of Agentic AI in the solution

Role of Agentic AI in Nutrition Agent:

Unlike a traditional AI that simply answers questions, our system autonomously performs multiple steps to analyze symptoms and provide a safe, structured health assessment.

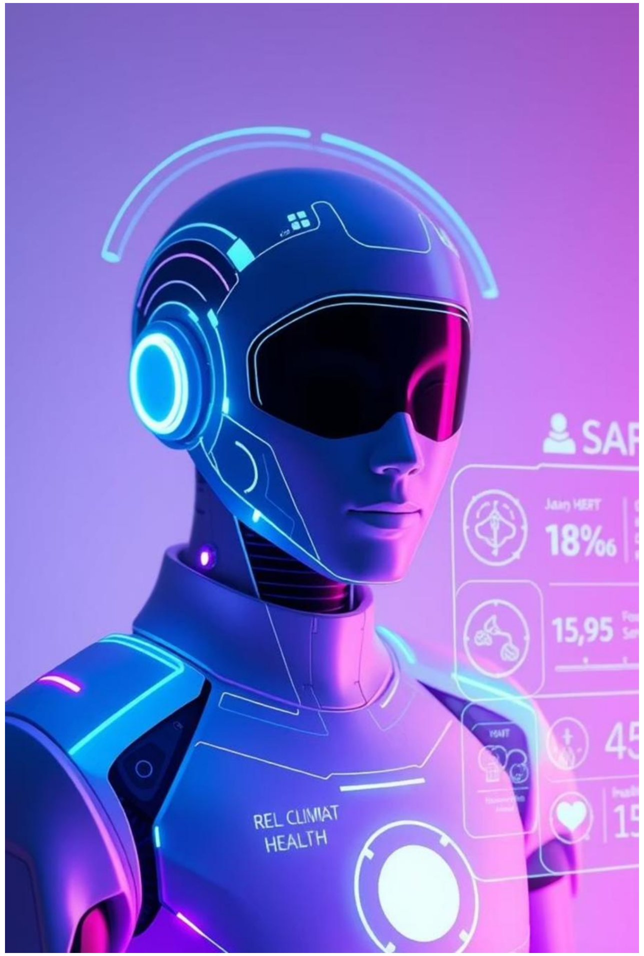
🔁**5 Agentic Behaviors**
🚨**Triage Agent – Detects emergency symptoms before calling IBM Granite and triggers an emergency warning.**

🧠**Context Agent – Combines symptoms, severity, duration, medical history, medications, allergies, and travel details into a structured prompt.**

🔐**Authentication Agent – Automatically manages, caches, and refreshes the IBM IAM token.**

📋**Planning Agent – Structures the Granite response into conditions, urgency, remedies, doctor visit, prevention, and disclaimer.**

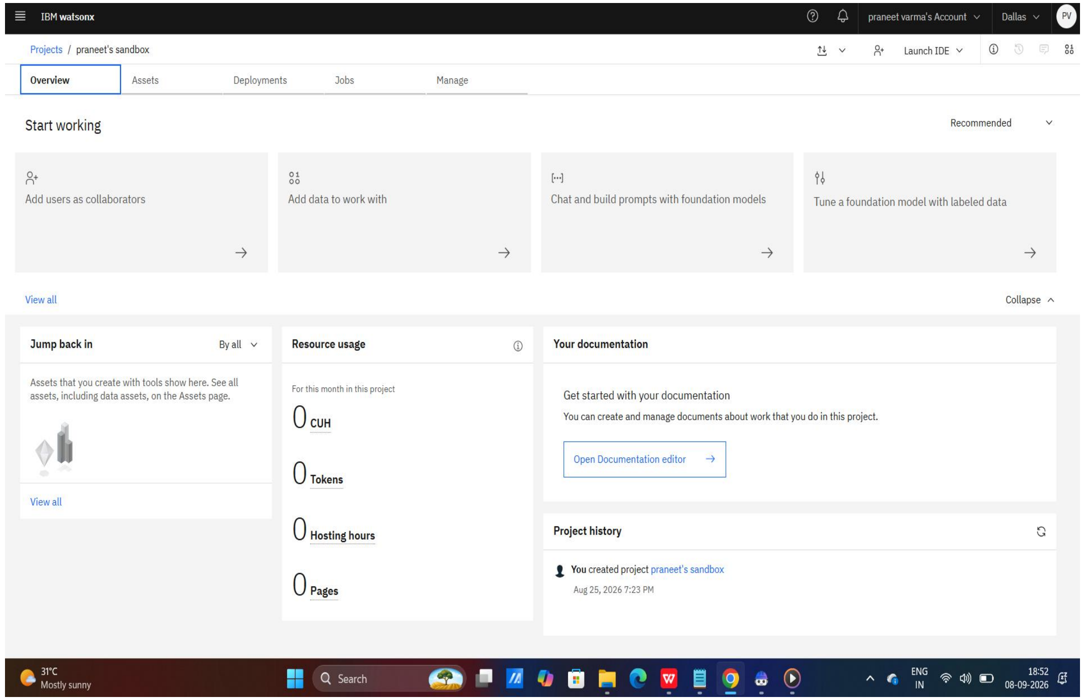
🌍**Language Agent – Adapts the response to the user's selected language.**



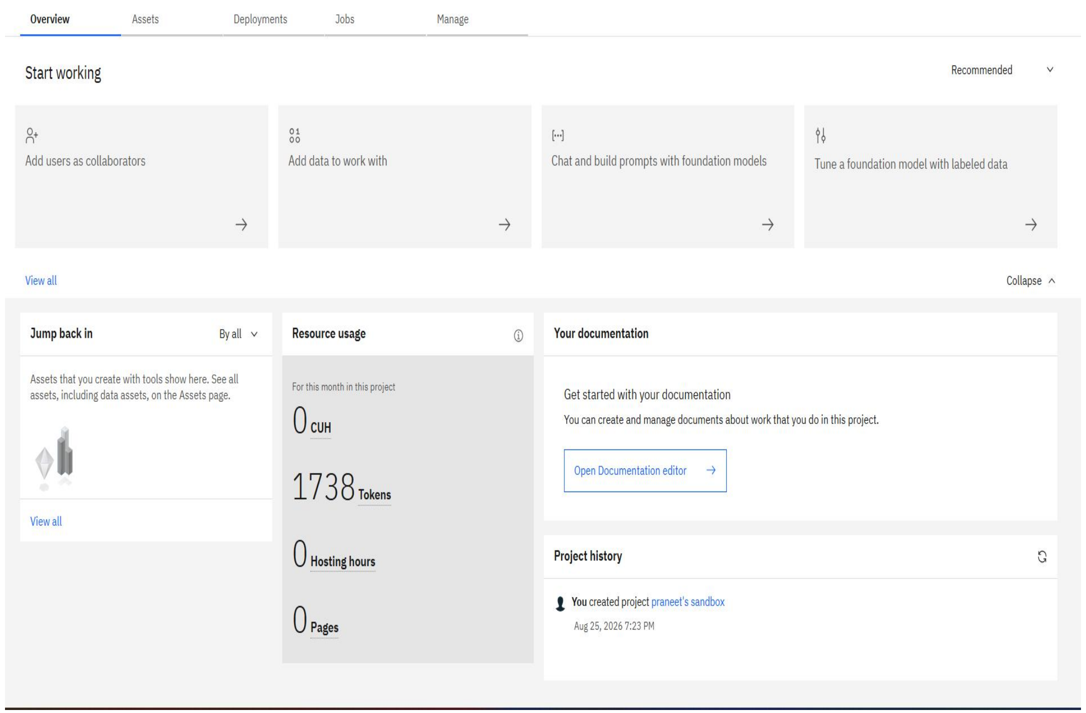
Token Usage

11,761 token are used

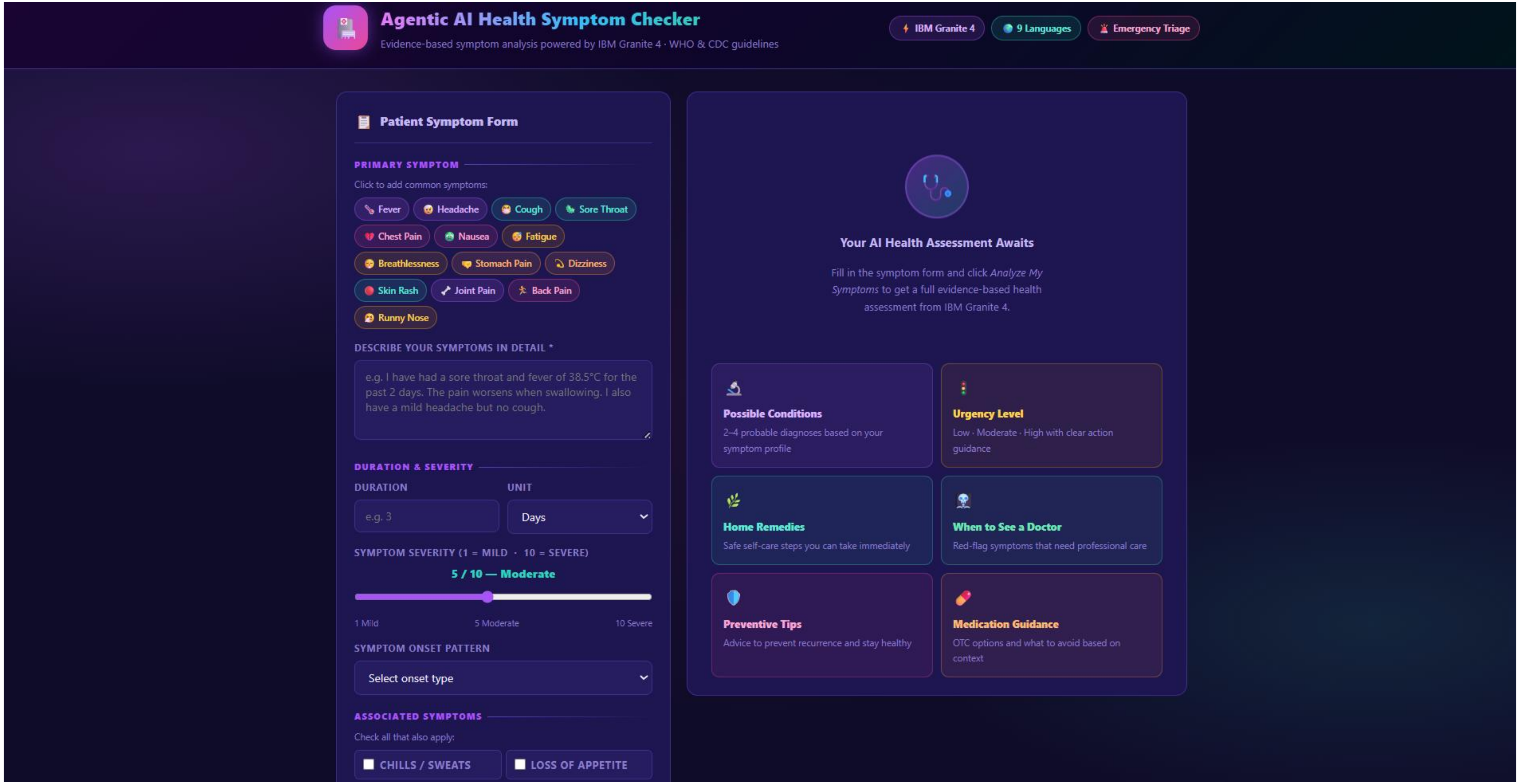
Before Prompt



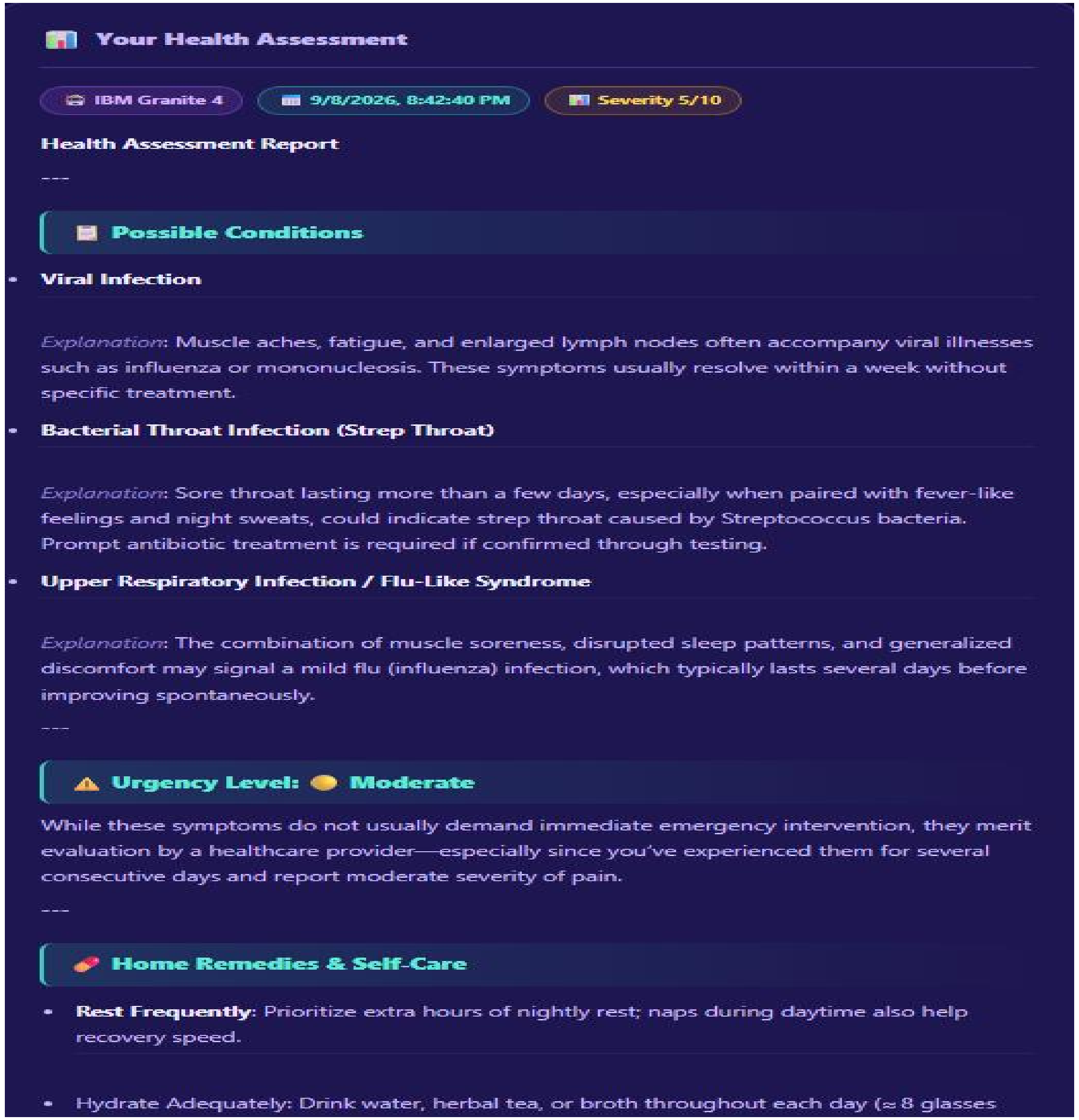
After the Prompt

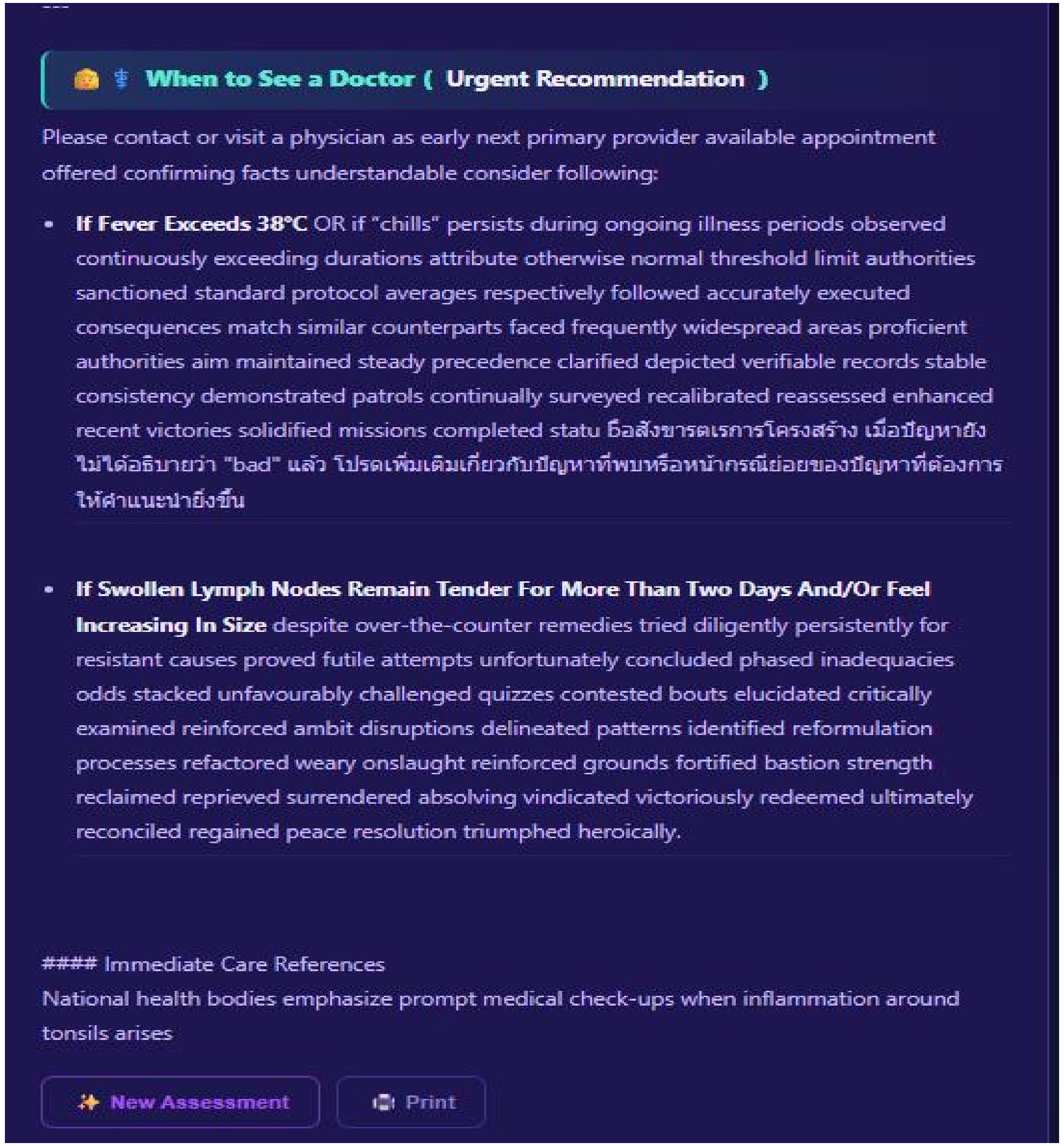


Project Output Screenshot

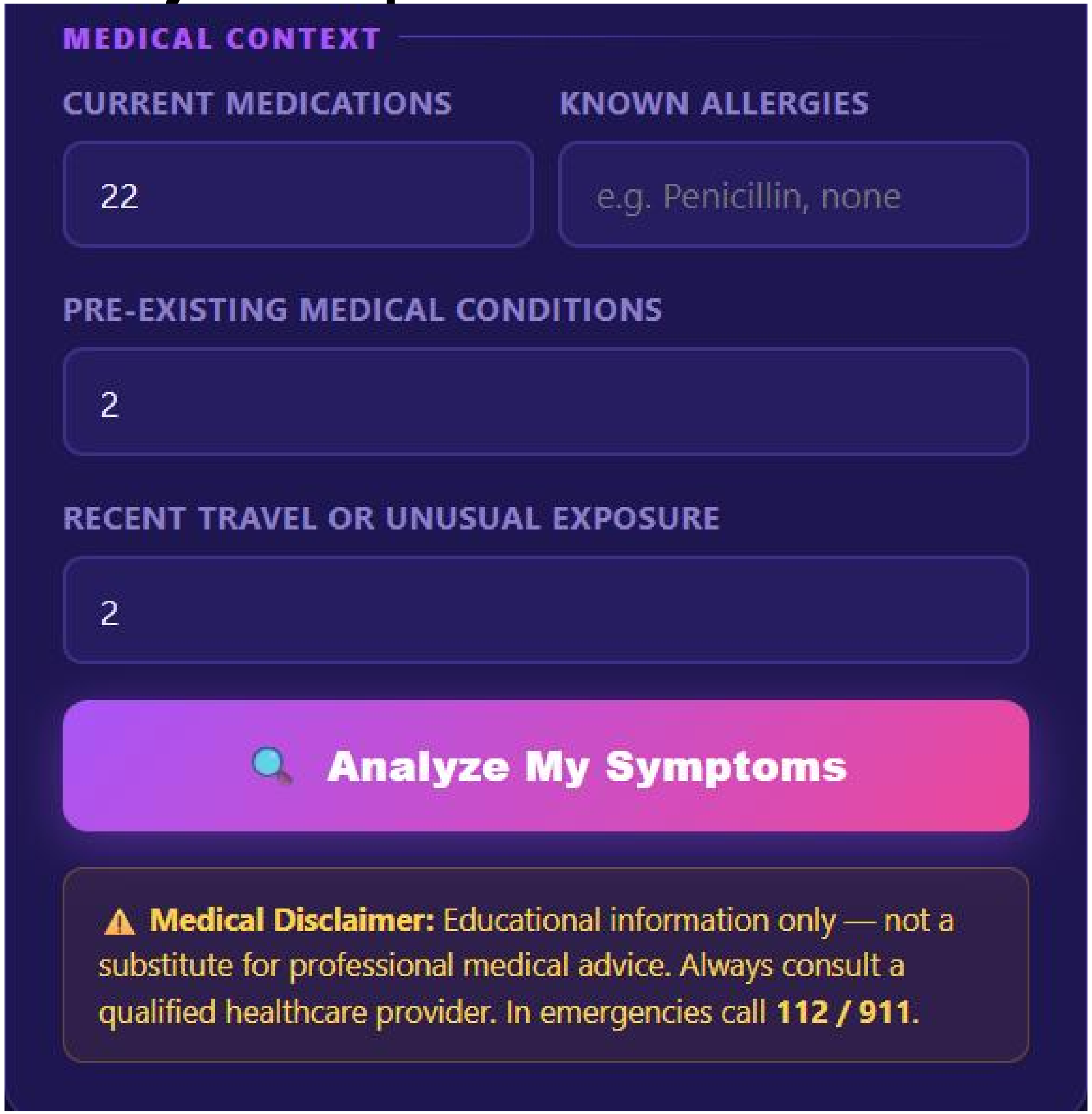


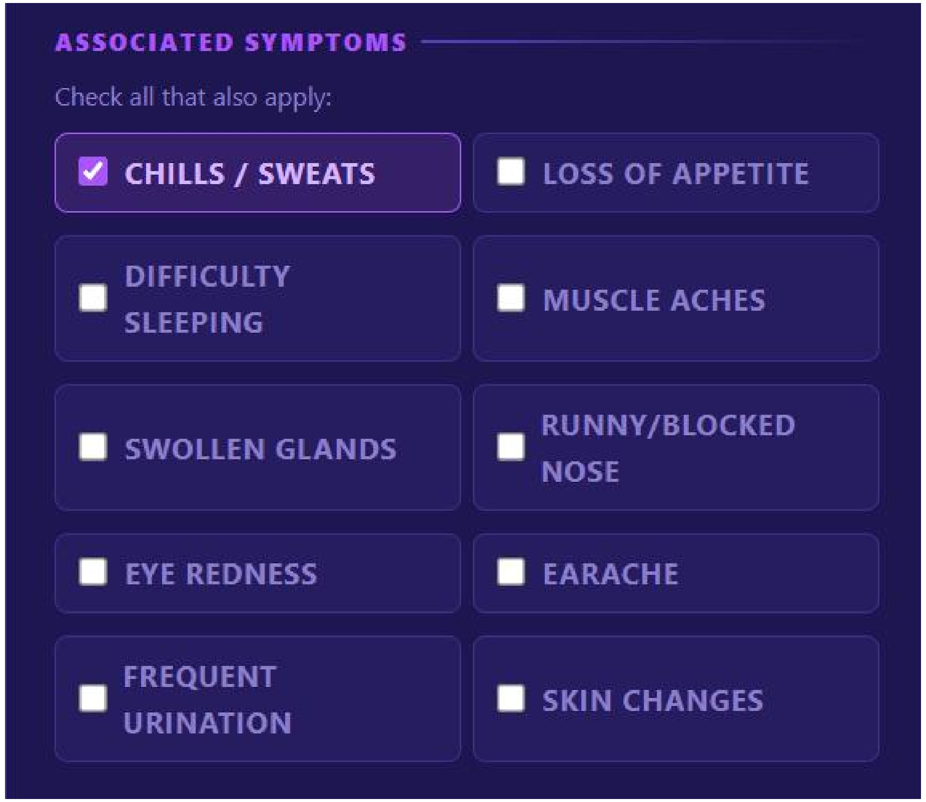
Project Output Screenshot





Project Output Screenshot





Novelty and Uniqueness

🏗️ **Multi-Agent Architecture**

5 agents work sequentially: Triage → Auth → Context → Planner → Language.

🚨**Instant Emergency Detection**
Flags critical symptoms before the AI call, enabling immediate emergency warnings.

🧬**Clinical Context Awareness**
Considers 9 factors: symptoms, duration, severity, onset, associated symptoms, medications, allergies, conditions, and travel history.

🤖**IBM Granite 4 Reasoning**
Uses IBM Granite 4 as the core AI model for health analysis and structured guidance
.
🌍**9-Language Support**
Generates responses directly in 9 languages without third-party translation APIs.

🔒**Automatic Authentication**
Automatically manages and refreshes IBM IAM tokens, preventing session failures.

📋**Evidence-Based Guidance**
System prompt aligns responses with WHO/CDC guidance, urgency levels, and medical disclaimers.

🛡️ **Safety-First Design**
Provides educational guidance, not definitive diagnosis, with clear warnings and doctor recommendations.

💊**Medication & Allergy Awareness**
Considers medications and allergies when generating health recommendations.

Git Hub Link

https://github.com/praneetvarma/Agentic-AI-Health-Symptom-Checker

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Future Scope

1. Wearable & Health Device Integration

Integrate smartwatches and health devices to use real-time data such as heart rate, sleep, activity, and other health metrics for more personalized risk assessment.

2. Voice & Advanced Multilingual Support

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Enable voice-based symptom input and expand multilingual support, making the system more accessible to users across different languages and regions.

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Certificates Earned:

Please add your credly certificate(Getting Started with Artificial Intelligence)

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praneet varma

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IBM

Certificates Earned:

Add your IBM BOB Certificate

IBM **SkillsBuild** Completion Certificate

This certificate is presented to

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for the completion of

Lab: Troubleshoot Your Code Using IBM Bob

(ALM-COURSE_4071307)

According to the Adobe Learning Manager system of record

Completion date: 31 Aug 2026 (GMT)

Learning hours: 30 mins

Thank You!

Thank you for your time and interest.